
Clash of Clans Base Archetype Classifier

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Colab Notebook: <https://tinyurl.com/4928ku8s>

GitHub Link: <https://github.com/SattiDaBeaver/APS360-Project>

Introduction

This project presents a "Clash of Clans Base Archetype Classifier", utilizing Convolutional Neural Networks (CNN) and Vision Transformers (ViT) to identify the archetype of the base from an image of the base layout. The project aims to assist attack strategy analysis and planning in both casual and competitive Clash of Clans (CoC) battles. A base consists of walls, defensive structures, and other buildings arranged in geometric layouts. An example of a CoC base is shown in Figure 1a.

In Clash of Clans, a game with a large monthly player base of 80 million players, specific attack strategies are stronger against certain base archetypes, such as box, ring, compartment, and island. Accurately identifying the base archetype provides a strategic edge to the attacker and can be very valuable, especially in competitive CoC matches. However, identifying the correct archetype can be non-trivial due to the visual complexity and limited time in preparing for attacks.

Deep learning is a suitable approach for this task as the archetype of the base is primarily defined by the geometric layout of the walls, rather than individual placement of other buildings. CNNs are good at identifying local features like wall patterns and compartments, while ViTs are good at identifying global features like the overall base structure. This combination allows the model to distinguish visually similar bases that differ in wall structure, making deep learning effective for this task.

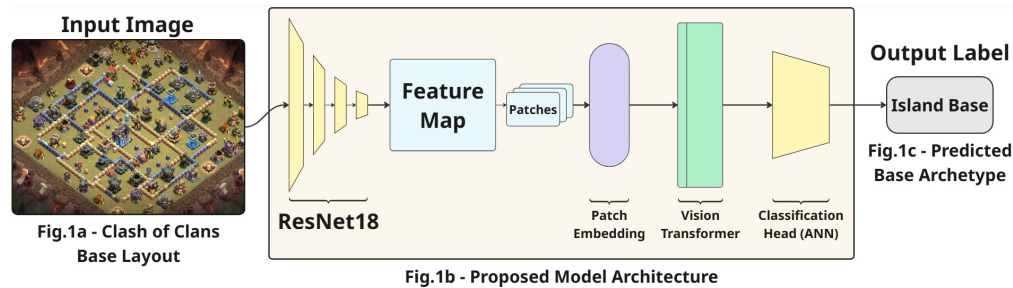


Figure 1: Proposed hybrid CNN-ViT model architecture

Background and Related Work

Listed below are some academic works that show why the CNN-ViT model is suitable for this task:

1. **ImageNet Classification with Deep CNNs (AlexNet)** [1]: Showed that deep CNNs are effective in learning spatial features from images, making CNNs a strong backbone for the project.
2. **Deep Residual Learning (ResNet)** [2]: Showed that ResNet can efficiently learn local features from residual connections. ResNet will be used as the CNN backbone for the model.
3. **Transformers for Image Recognition at Scale** [3]: Showed that Vision Transformers (ViT) capture global image structure and are highly accurate for image classification.
4. **Introducing Convolutions into Vision Transformers** [4]: Showed that combining ViTs with a CNN backbone increases performance and efficiency, making the architecture ideal for the project.
5. **Early Convolutions Help Transformers See Better** [5]: Showed that a lightweight convolutional stem for the ViT models is a more robust architecture than standalone ViTs.

Data Processing

Images were collected for Town Hall Level 18 from two public Clash of Clans base sharing websites: Clash Champs [6] and CoC Layouts [7], using a custom Python web scraper run locally. The script uses Selenium and BeautifulSoup for image parsing, and curl for downloading images. Most images from Clash Champs are 1920×872 px (varying slightly by base), while images from CoC Layouts are 1000×460 px. In total around 2650 images were collected from the websites.

Upon inspection, most images from Clash Champs were mislabeled. Hence, images were manually labeled while removing garbage images, giving a total of 2623 labeled images. The four archetypes are: **Box** (large outer walled section); **Ring** (thick outer walls); **Compartment** (interconnected walled sections); and **Island** (multiple isolated pockets). The class distribution is shown in Table 1.

Each image was resized to 300px on the shorter side, center cropped to 256×350 px to remove borders, then resized to 224×224 px for ResNet compatibility. During training, the images are randomly horizontally flipped for augmentation, effectively doubling the initial dataset size. To address class imbalance, a weighted random sampler was used. An example processed image is shown in Figure 2.

Class	Train	Val+Test	Total
Box	466	117	583
Comp.	893	223	1116
Island	418	105	523
Ring	321	80	401

Table 1: Class distribution (80/10/10 split)



Figure 2: Left: raw image; Right: processed image

Baseline Model Architecture and Results

An Artificial Neural Network (ANN) was used as the baseline. The processed images are flattened to a 1D tensor of size $224 \times 224 \times 3 = 150,528$, and passed through four fully connected layers (512, 256, 64, and 4) with ReLU activations. The output is a 4-class vector trained using cross-entropy loss. The model was trained for 30 epochs with batch size 32, learning rate 0.001, and Adam optimizer. The total number of parameters is ~ 77.2 M. The baseline architecture is shown in Figure 3.

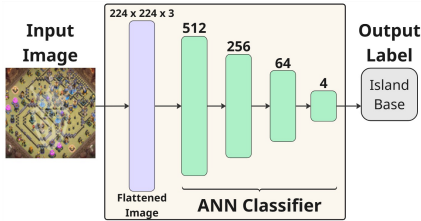


Figure 3: ANN baseline architecture



Figure 4: ANN train/validation curves

The ANN achieved a best validation accuracy of **40.8%** (0.592 error), compared to a random chance of 25%. Over the epochs, the training loss decreases steadily, however the validation loss increases after the first few epochs indicating overfitting. This shows that the ANN does not generalize very well. The training and validation loss/error curves are shown in Figure 4.

Final Model Architecture

The final model is a hybrid CNN-ViT architecture. A ResNet18 backbone (pretrained on ImageNet) extracts local geometric features (like wall compartments), which are then processed by a Vision Transformer (ViT) to capture global structural patterns (like base structure). The output is passed to a classification head for base archetype classification. The final architecture is shown in Figure 1.

The ResNet18 backbone extracts the feature maps ($512 \times 7 \times 7$) from the input image. The backbone has all layers frozen except the last layer (layer 4), which is fine-tuned. The 7×7 feature maps are treated as a sequence of patch tokens, each with 512-dimensional feature vector. These are projected to a 128-dimensional vector via a linear layer, giving a sequence of 49 patch embeddings.

A trainable CLS (classification) token is prepended to the patch tokens, and trainable positional encodings are added to all tokens. The sequence is passed through a 2-layer Transformer encoder with 4 attention heads, feedforward dimension of 512, and dropout of 0.1. The transformer output is passed to a classification head: Linear(128, 64) → ReLU → Dropout(0.4) → Linear(64, 4). The frozen backbone has $\sim 2.8\text{M}$ parameters, the fine-tuned layer4 has $\sim 8.4\text{M}$ parameters, while the trainable components (projection, transformer, classifier) have $\sim 470\text{K}$ parameters (11.7M total parameters).

Quantitative Results

The CNN-ViT model was trained for 30 epochs with a batch size of 32 and the Adam optimizer. layer4 of the pre-trained ResNet18 backbone was fine-tuned with a learning rate of 10^{-6} , while all other trainable parameters used a learning rate of 10^{-4} . The best validation accuracy of **57.6%** (0.424 error) was achieved at epoch 20, after which the model begins to overfit as the training loss continues to decrease while validation loss diverges. The CNN-ViT outperforms the ANN baseline (40.8%), demonstrating that the hybrid architecture captures spatial and global structural features that a flat ANN cannot. The training and validation loss/error curves are shown in Figure 5.

Table 2 shows the per-class validation accuracy. This shows that Ring and Compartment are classified most reliably, while Island is the most difficult to classify.

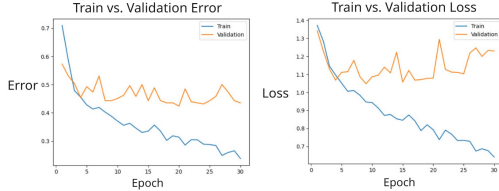


Figure 5: CNN-ViT train/validation curves

Class	Val Accuracy
Box	51.9%
Comp.	60.8%
Island	47.5%
Ring	70.2%

Table 2: Per-class validation accuracy of the CNN-ViT model

Qualitative Results

Figure 6 shows three predictions made by the model: confidently correct (left), wrong but close (middle), and confidently wrong (right). Figure 7 shows the validation confusion matrix.



Figure 6: Sample predictions

	Box	Comp	Island	Ring
Box	27	15	4	6
Comp	18	62	18	4
Island	8	22	29	2
Ring	4	4	6	33
	Box	Comp	Island	Ring

Figure 7: Val confusion matrix

The Ring base was classified correctly with 97% confidence, which is consistent with the confusion matrix showing Ring is the most reliably classified archetype (70.2% validation accuracy). Intuitively, the Ring bases have a thick outer wall structure which is easy to identify in general as well.

The Compartment base was misclassified as Box with 54% confidence, with Compartment as the second highest at 40.4%. In general, evident from the confusion matrix, the model frequently confuses between Box and Compartment, likely due to many bases exhibiting features of both base archetypes.

The Island base was misclassified as Compartment (37.7%), with the true label of Island being much lower at (16.7%). As evident from the confusion matrix, the model often misclassifies Island as Compartment, likely due to both consisting of multiple separated walled sections.

Model Evaluation on New Data

A held-out test set (10% of the dataset), which was not used during training or validation and did not influence hyperparameter tuning, was used to evaluate the generalization of the model. The CNN-ViT model achieved a test accuracy of **62%**, which is comparable to the validation accuracy (57.6%), indicating good generalization and minimal overfitting.

The model correctly classified most samples, but struggled with visually similar archetypes (e.g., Compartment vs Box), consistent with observations from the validation dataset. To further evaluate performance on real-world data, 6 samples were collected from personal in-game screenshots. These images provide a qualitative assessment of how the model behaves on unseen real-world data (e.g., what patterns does the model pick up on). Figure 8 shows samples from in-game screenshots, and Figure 9 shows the test confusion matrix.



Figure 8: Samples from in-game screenshots

True Label	Box	34	8	8	4
	Comp	15	74	15	9
	Island	8	15	28	3
	Ring	5	4	6	27
		Box	Comp	Island	Ring
		Predicted Label			

Figure 9: Test confusion matrix

Overall, the model generalizes well to new data, with acceptable performance considering the visual similarities between archetypes, and several bases exhibiting properties of multiple archetypes. Given the inherent ambiguity of the task and the dataset size, 62% test accuracy is a reasonable result.

Discussion

The results show a clear progression from the ANN baseline (40.8% val accuracy) to the CNN-ViT model (57.6% val accuracy), confirming that spatial feature extraction is necessary for this task. An ANN cannot capture geometric wall features, while the ResNet18 (CNN) backbone provides meaningful local feature extraction, and the custom trained ViT provides global feature attention.

Qualitatively, the model identifies the intended features reasonably well. Ring bases have the highest classification accuracy, likely due to their visually distinct thick walls and hollow center. Compartment bases are also identified well, though the model tends to over-predict this class, likely due to its over-representation in the dataset. Box and Island are the hardest classes, as many bases exhibit characteristics of multiple archetypes, for example a base can have a large outer perimeter (Box) while also having internal walled sections (Compartment), making the classification inherently ambiguous.

This ambiguity is a fundamental limitation of the problem, as base archetypes are not strictly one-hot. A base may belong to multiple classes, and the ground truth labels also reflect subjective human choice during manual labeling. This introduces inconsistency that the model cannot fully overcome.

A key finding was that the CNN-ViT model significantly outperforms the ANN baseline, despite having far fewer trainable parameters. The dataset size (2623) and class imbalance (compartment overrepresented at $\sim 2\times$) also limit performance. Despite the limitations, the model demonstrates that a CNN-ViT model is an effective approach for base archetype classification, and that the performance is bottlenecked more by dataset quality and problem ambiguity, than model architecture at this scale.

Ethical Considerations

While the model is intended to assist players in planning and analyzing attack strategies, it should not completely replace human decision making. In competitive Clash of Clans tournaments, use of such classification tools could give the user an unfair advantage and may violate competition rules.

The dataset is collected from public base sharing websites which are biased towards popular competitive-style base designs. This could hinder the model’s ability to classify novel and less common base archetypes. No personal information is collected from the data sources, only the base layouts and labels are collected. Therefore, the model poses minimal privacy risks.

References

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